

Project Report: Inventory Management System

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1. Project Overview

This report summarizes the development of an internal inventory management system built to replace a legacy spreadsheet-based workflow. The system allows warehouse staff to track stock levels, manage purchase orders, and generate monthly reports through a web-based interface.

2. Objectives

- Automate stock level tracking and low-inventory alerts.
- Integrate with the existing ERP system via REST API.
- Reduce manual data entry errors by 80 percent.
- Provide a dashboard for real-time inventory visibility.

3. Technology Stack

Component	Technology
Backend	Python / FastAPI
Database	PostgreSQL 15
Frontend	React 18
Deployment	Docker / AWS EC2
Authentication	OAuth 2.0 / JWT

4. Project Timeline

Phase	Duration	Status
Requirements & Design	2 weeks	Completed
Backend Development	4 weeks	Completed
Frontend Development	3 weeks	Completed
QA & Testing	2 weeks	Completed
Deployment & Handover	1 week	In Progress

5. Results and Outcomes

The system was successfully deployed to the staging environment in April 2026. Initial testing showed a 92 percent reduction in manual data entry errors compared to the previous workflow. Average page load time is under 300ms. The ERP integration was validated with 100 percent data parity across 3,000 test records.

6. Conclusion

The project was delivered on schedule and within budget. Full production rollout is planned for June 2026 following final user acceptance testing. Documentation and training materials have been prepared for warehouse staff.